Making blockmodeling easier

R Shiny as a GUI for generalised blockmodeling

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Blockmodeling

Definition:	Given a definition of nodes' 'equivalence', blockmodeling forms the 'best' clusters of equivalent units
Problem:	Although there are some advanced tools for blockmodeling, they can be difficult to use for novices and social scientists
Research puzzle:	Finding a balance between user-friendliness and the high number of parameters and options for blockmodeling
Aim/Goal:	Creating an easy-to-use graphic user interface (GUI) for blockmodeling based on the R package `blockmodeling`
Method:	R Shiny, an R package for web apps deployable within interactive webpages, documents, and standalone software

Why to use it?

Deploy	Does not require an installation of R
Fast start	The app has virtually no learning curve
Progress	Using the app can be a first step towards learning how to block-model by coding in R

Who is it for?

Use contexts

- Graduate classes on quantitative methods in social sciences (with/out R)
- Quantitative and mixed-method research on policy networks, financial networks

Supported inputs

Text files:	All the most common matrix formats are supported: adjacency matrix; edge list + node list, incidence matrix
Pajek files:	For compatibility with Pajek, the app supports `.mat` (adj. matrix + node list) and `.net` files (edge list + node list)

QR Corner

Scan me!

The app is on **CRAN**

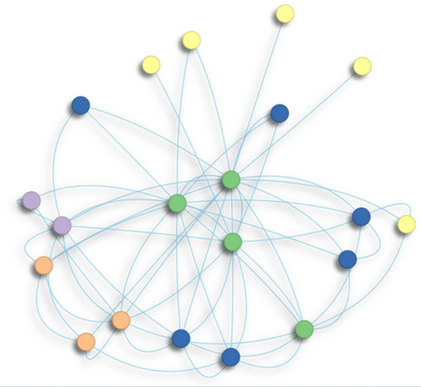
The R package `blockmodeling` is on CRAN

Blockmodeling options

Base options:	Full support for the approaches (binary, valued, etc.) and the block-types (complete, null, functional) available in the R package `blockmodeling`.
Advanced options:	Interface support for building pre-specified block-models and importing them from R files (`.RDS`)
Store and restore	Importing and exporting blockmodeling results from/to `.RDS` files + Restoring up to 10 results in a session

The app's outputs

Networks info:	General information on the uploaded network from the package `network`
Block-modeling:	Two textual summaries, image matrix, error matrix, original and partitioned matrix plots.
Plots:	Fully customisable plots produced using the packages `network` (simple, static), `igraph` (more advanced, static) and `visNetwork` (interactive)
Pajek files:	For compatibility with Pajek, the app can export the partitions as `.clu` files
Other files:	The image matrix and the blockmodeling's results can be exported as text or `.RDS` files



	SW	SCW	SSR	JSWE	CYSR	CCQ	CW	PW	CAN	JGSW	BJSW	JSP	AMH	IJSW	SWRA	ASW	CSWJ	SWHC	SWG	FR
SW		6	5	3			5	2							1	2		4	2	1
SCW	12		4	2			2		1	1					2	1	1	1	1	1
SSR	11	3		1			2								2	1				
JSWE	6	2	2												2	1		1		
CYSR	3	1	1			0	7	1	1						0					0
CCQ		0					1													
CW	6	3	1	1	1			1	1											
PW							0		1											
CAN	1	1					1													
JGSW	2	2																		
BJSW	2			1																
JSP			1																	
AMH	0																			
IJSW	0																			
SWRA	4	1	4	2			1													
ASW	7		2	2				1							1					
CSWJ	4	5	2																	
SWHC	3	2																		
SWG	4	1	1	1							0							1		
FR	1	2																		

* all values in cells were multiplied by 0.1

Conclusion

- The app makes generalised blockmodeling and some basic network analysis in R easier to execute and understand than using more traditional means;
- Despite an inevitable trade-off between ease of use and availability of numerous option, the app includes many options – including some rather advanced ones;
- The app's flexibility and ability to integrate with other R packages and with Pajek make it a useful tool for social-science researchers from diverse backgrounds;
- The app provides an accessible tool to get acquainted with blockmodeling, making it a good educational tool as well,

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